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Calibration apparatus for offset vehicle sensor

Abstract

A calibration apparatus and method suitable for calibration of an offset sensor of a subject vehicle. The calibration apparatus comprises a reference structure that is placed into, a reference locus using image data generated by a camera associated with the reference structure. An offset-target structure is then placed into position by coupling the offset-target structure to the reference structure, the coupling providing an appropriate locus for the offset-target structure during calibration of the offset sensor. The coupling restricts the linear and rotational displacement of the offset-target structure during calibration.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to the calibration of vehicular sensors and an associated apparatus used for the calibration thereof.

BACKGROUND

(2) Advanced driver assistance (ADAS) functions of modern vehicles are becoming more robust and complex. ADAS functions of vehicles rely upon an array of sensors arranged around the vehicle to monitor the environmental and traffic conditions the vehicle is subjected to. Calibration of these sensors is critical to ensure proper and safe operation of the ADAS functions of the associated vehicle. Specialized calibration apparatuses have been utilized to provide calibration of the sensors associated with ADAS functions.

(3) Existing calibration apparatuses are often bulky, unwieldy, and rely upon controlled environments and repeated measurements with respect to the subject vehicle to find their appropriate placement for a successful calibration. It would be desirable for a calibration apparatus

that was functional with a minimized number of measurements in order to expedite the setup process and streamline calibration, even in field environments.

SUMMARY

(4) One aspect of this disclosure is directed to a calibration apparatus for an offset sensor of a subject vehicle. The apparatus may comprise a reference structure having a horizontal member, the horizontal member disposed in a direction substantially parallel with a horizontal plane, a camera configured to be coupled to the reference structure and operable to generate image data depicting the subject vehicle, and an offset-target structure having a guide arm and configured to be detachably couple to the horizontal member such that the guide arm interfaces with the horizontal member. The calibration apparatus may be configured for a calibration procedure of the offset sensor when the wheel-target structure is coupled to a wheel of the subject vehicle, the reference structure is positioned relative to the subject vehicle based upon image data generated by the camera, and the offset-target structure is coupled to the horizontal member. Some embodiments may additionally comprise a wheel-target structure configured to detachably couple to a wheel of the subject vehicle.

(5) Another aspect of this disclosure is directed to a method for placing an offset-target structure into a locus suitable for calibration of an offset sensor of a subject vehicle, the offset-target structure having a guide arm. The method comprises positioning a reference structure into a reference locus and coupling the offset-target structure to the reference structure such that the guide arm interfaces with the horizontal member. In some embodiments, the method may additionally comprise detachably coupling a wheel-target structure to a wheel of the subject vehicle. The reference structure may have a horizontal member and a camera. The guide arm restricts the motion of the offset-target structure with respect to pitch, yaw, or roll when the offset-target structure is coupled to the reference structure. The reference locus may be defined in relation to the subject vehicle. In the method, the reference locus is found using image data generated by the camera while the wheel-target structure is coupled to the wheel.

(6) The above aspects of this disclosure and other aspects will be explained in greater detail below with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an illustration of a subject vehicle during a calibration procedure for an offset sensor of the subject vehicle.

(2) FIG. 2 is an illustration of a reference structure and an offset-target structure of a calibration apparatus for an offset sensor of a vehicle.

(3) FIG. 3 is an illustration of a wheel-target structure of a calibration apparatus for an offset sensor of a vehicle.

(4) FIG. 4 is an illustration of a diagnostic device illustrating image data on a display along with a silhouette reference template.

(5) FIG. 5 is a view of a disassembled offset-target structure of a calibration apparatus for an offset sensor of a vehicle.

(6) FIG. 6 is a closeup view of a guide arm and horizontal member of a calibration apparatus for an offset sensor of a vehicle.

(7) FIG. 7 is a flowchart of a method for arranging a calibration apparatus such that it is suitable for calibrating an offset sensor of a vehicle.

DETAILED DESCRIPTION

(8) The illustrated embodiments are disclosed with reference to the drawings. However, it is to be understood that the disclosed embodiments are intended to be merely examples that may be

embodied in various and alternative forms. The figures are not necessarily to scale and some features may be exaggerated or minimized to show details of particular components. The specific structural and functional details disclosed are not to be interpreted as limiting, but as a representative basis for teaching one skilled in the art how to practice the disclosed concepts.

(9) FIG. 1 depicts a vehicle **100** acting as a subject vehicle of a calibration procedure directed to an offset sensor **102** associated with the vehicle. In the depicted embodiment, vehicle **100** may comprise additional sensors in addition to offset sensor **102**. The sensors available to vehicle **100** may comprise front, rear, or side sensors arranged to detect objects or environmental conditions in a direction perpendicular or parallel to the normal forward motion of vehicle **100**. In contrast, offset sensor **102** may be configured to detect objects or environmental conditions at oblique angles with respect to the normal forward motion of vehicle **100** or the perpendicular direction thereto.

Examples of offset sensors may comprise lane-change sensors, blind-spot sensors, lane detection sensors, or any other offset sensor known to one of ordinary skill in the art without deviating from the teachings disclosed herein. In the depicted embodiment, offset sensor **102** comprises a lane-change sensor, but other embodiments may comprise additional or different offset sensors without deviating from the teachings disclosed herein. The sensors of vehicle **100**, such as offset sensor **102**, may comprise known types, such as image sensors, camera sensors, optical sensors, radar sensors, lidar sensors, proximity sensors, motion sensors, or any other vehicular sensor known to one of ordinary skill in the art without deviating from the teachings disclosed herein. By way of example, and not limitation, the depicted embodiment comprises a configuration of offset sensor **102** comprises an optical sensor, but other embodiments may comprise other configurations without deviating from the teachings disclosed herein.

(10) Vehicle **100** additionally comprises a number of wheels **104**. In previous calibration techniques, placement of the wheels were utilized with respect to another calibration apparatus. In the depicted embodiment, each of the components of the calibration apparatus may positioned into loci suitable for calibration in relation to the vehicle. Advantageously, placement of the apparatus in relation to the vehicle permits a relatively arbitrary placement of the vehicle to be supported by the calibration apparatus. In an additional advantage, the various components of the calibration apparatus are simply and easier to move manually than vehicle **100**, which typically must be driven into a particular position.

(11) The calibration apparatus comprises a reference structure **105** having a horizontal member **107** and camera **109**, a wheel-target structure **111**, and an offset-target structure **113** having a guide arm **115**. In the depicted embodiment, the camera **109** generates image data that is utilized to determine a reference locus for the reference structure **105**. Wheel-target structure **111** is additionally utilized to provide a distinctive subset of image data to be utilized in the placement of reference structure **105**. When the reference structure **105** is successfully positioned at its respective reference locus, offset-target structure **113** may be coupled to reference structure **105** by detachably coupling guide arm **115** and horizontal member **107**. Offset-target structure **113** is designed according to a specification such that when it is coupled to reference structure **105**, offset-target structure **113** is in a suitable offset locus for the calibration of offset sensor **102**. Notably, the placement of reference structure **105** may utilized for the calibration of other sensors of vehicle **100**, but its placement at the reference locus may be utilized to simply and efficiently place offset-target structure **113** without additional measurements beyond those already undertaken during placement of reference structure **105**. This advantageously minimizes the setup requirements for the calibration apparatus. In the depicted embodiment, wheel-target structure **111** is mounted to rear wheel **104b**, but other embodiments may utilize different mounting, such as to front wheel **104a**. In other embodiments, the calibration apparatus may omit wheel-target structure **111** without deviating from the teachings disclosed herein. In the depicted embodiment, the calibration apparatus comprises a single reference structure **105** and a single offset-target structure **113**, but other embodiments may comprise a different number of each of these components without deviating from the teachings

disclosed herein. By way of example, and not limitation, an embodiment may comprise a plurality of offset-target structures **113**, each of the offset-target structures placed on opposite ends of reference structure **105** in order to provide calibration targets for symmetrical offset sensors **102** on either side of vehicle **100**.

(12) FIG. 2 depicts a view of reference structure **105** and offset-target structure **113** prior to detachable coupling of the two components. In the depicted embodiment, camera **109** is mounted near the top reference structure **105** and angled such that a downward view of a subject vehicle (e.g., vehicle **100**, see FIG. 1) is acquired in the image data generated by camera **109**. The generated image data may be sent to a display (not shown) and updated in real time as reference structure **105** is moved in relation to its subject vehicle. Use of the real-time update of image data advantageously permits a user to find the proper reference locus for reference structure **105** with a minimal number of measurements between reference structure **105** and its respective subject vehicle. When the reference structure **105** has been positioned into the appropriate reference locus, the depicted embodiment may comprise wheel locks suitable to temporarily lock the reference structure **105** into position at the reference locus.

(13) In the depicted embodiment, horizontal member **107** is arranged on reference structure **105** near the bottom of the structure. In the depicted embodiment, horizontal member **107** may house a number of secondary cameras **209** that can be used to enhance, supplement, or replace the image data generated by camera **109**. In the depicted embodiment, the placement of horizontal member **107** and arrangement of secondary cameras **209** is optimized to generate image data depicting wheel-target structure **111** (see FIG. 1). In the depicted embodiment, wheel-target structure **111** may be utilized to enhance the precision with which the reference structure **105** can be positioned at an appropriate reference locus. Some embodiments may not utilize one or more of secondary cameras **209** without deviating from the teachings disclosed herein.

(14) Offset-target structure **113** is additionally shown to comprise a detachable target **213**. Detachable target **213** may comprise one of several designs suitable for calibrating an offset sensor of different specifications. Advantageously, detachable target **213** may be detached from offset-target structure **113** and replaced with an alternatively-configured detachable target **213** to enhance the compatibility of the calibration apparatus with a greater variety of sensor specifications. In some embodiments, detachable target **213** may be constructed of desirable materials that optimize compatibility with a particular design of offset sensor. Detachable target **213** may comprise a rigid construction, such as metal, wood, or rigid plastics without deviating from the teachings disclosed herein. In some embodiments, detachable target **213** may comprise a malleable construction, such as textile fabric, plastic sheets, or polymers without deviating from the teachings disclosed herein. In the depicted embodiment, by way of example and not limitation, detachable target **213** comprises a vinyl mat target, which advantageously provides to a user a lightweight and portable target for the calibration apparatus. A vinyl mat target advantageously provides for an easy replication of a specified calibration design without sacrificing desired ruggedness for long-term use of the detachable target **213**. A vinyl mat target additionally advantageously permits a user to easily store detachable target **213** when detached from offset-target structure **113** by rolling or folding the target when not in use. This ease of use and storage enhances the portability and flexibility in using the calibration apparatus, while also reducing costs for detachable target **113**.

(15) FIG. 3 is an illustration of wheel-target structure **111**. Wheel-target structure **111** comprises a mount handle **301** suitable for a user to carry or position the wheel-target structure **111** when not in use, but also suitable to suspend the wheel-target structure **111** over a wheel of a vehicle While setting up a calibration procedure. Wheel-target structure **111** additionally comprises a number of mount braces **303** connected to each other and to mount handle **301** via a brace tie **305**. Mount braces **303** rest against the wheel or the vehicle itself to properly orient wheel-target structure **111** in a suitable position with respect to yaw, pitch, roll, and tilt to ensure that wheel-target structure **111** acts as a suitable reference while positioning reference structure **105** (not shown; see FIG. 1,

FIG. 2). In the depicted embodiment, wheel-target structure **111** additionally comprises a height adjustment mechanism **307** operable to configure the wheel-target structure **111** such that it is compatible with a variety of wheels of different sizes and configurations. When configured into a specified arrangement, wheel-target structure **111** is designed to position a wheel target **309** into a position suitable for use as a reference when positioning reference structure **105**. In some embodiments, wheel target **309** may be detachable, and may be suitably replaced with an alternatively-configured wheel target **309** that is designed to be compatible with a different sensor model, vehicle model, or a different particular reference locus. Some embodiments of the calibration apparatus will utilize two or more wheel-target structures **111** without deviating from the teachings disclosed herein. Some embodiments of the calibration apparatus will not utilize a wheel-target structure **111** without deviating from the teachings disclosed herein.

(16) FIG. **4** is a depiction of a diagnostic device **400** in data communication with camera **109** of an associated reference structure **105** (not shown; see FIG. **1**, FIG. **2**). Some embodiments may additionally be in data communication with secondary cameras **209** without deviating from the teachings disclosed herein. Diagnostic device **400** comprises a display **401** that displays image data **403** generated by the associated camera in real time. In the depicted embodiment, the image data **403** depicts a subject vehicle (such as vehicle **100**; see FIG. **1**) arranged with a pair of wheel-target structures **111** (see FIG. **3**). Also displayed is a reference template **405** depicting an expected aspect of the image data to be observed when the associated reference structure **105** has been properly positioned at the reference locus. In the depicted embodiment, the reference template **405** comprises a silhouette reference template depicting an upper silhouette of the vehicle and the wheel-target structures, but other embodiments may comprise other reference templates. Other embodiments having different reference templates may comprise full depictions of the vehicle, partial depictions of the vehicle, an arrangement of symbols on the display, an active positional/direction indicator guiding a user as to how to move reference structure **105**, or any other displayable reference template known to one of ordinary skill in the art without deviating from the teachings disclosed herein. In some embodiments, diagnostic device **400** may be mounted to a designated placement (not shown) on reference structure **105** in order to optimize the ergonomics of placement. For example, diagnostic device **400** may be mounted to a placement position on the side of reference structure **105** opposite camera **109** so that a user may watch the display **401** in real time while positioning the reference structure **105** without interfering with the image data generated by camera **109**. In the depicted embodiment, diagnostic device **400** comprises a tablet processing device having a processor (not shown) and a touchscreen display **401**. In some embodiments, diagnostic device **400** may comprise a smartphone, portable computing device, wearable computing device, or specialized camera display without deviating from the teachings disclosed herein. In some embodiments, diagnostic device **400** may comprise additional user input and output functions, such as auditory or haptic controls or output components, without deviating from the teachings disclosed herein.

(17) Diagnostic device **400** may comprise a memory (not shown) storing thereon a plurality of reference templates to advantageously improve compatibility of the calibration apparatus with a variety of models of sensor or models of vehicle. In some such embodiments, multiple reference templates having different display configurations—such as a silhouette reference template as well as a full-image reference template—may be provided for a single configuration of a subject vehicle in order to accommodate user display preferences. In some embodiments, additional reference templates may be accessed from external storage, from a network connection, from an Internet connection, or from a cloud-based storage device without deviating from the teachings disclosed herein. Accessing additional, reference templates from external locations advantageously improves the usability of diagnostic device **400** by providing, compatibility with new and different models of sensor or vehicle as they become available. Access to these additional reference templates may be provided for a charge, such as a flat fee for external storage media or a subscription service for

continued access to cloud storage.

(18) FIG. 5 depicts a disassembled view of offset-target structure **111**. In this depiction, detachable target **213** has been removed from the target frame **501**. It is additionally shown that guide arm **115** is configured to be detachably coupled to target frame **501** using a coupling placement **503**. When guide arm **115** is coupled to coupling placement **503a** it is in suitable location to interface with a horizontal member **107** of a reference structure **105** (see FIG. 1) for calibration of an offset sensor. However, this placement of guide arm **115** may not be desired for portability or storage of the offset-target structure **111**. When a more compact arrangement is desired, guide arm **115** may instead be detachably coupled to coupling placement **503b**, located within target frame **501**. Coupling placement **503b** may advantageously provide a place to detachably couple guide arm **115** that is less awkward for storage or portability of the calibration apparatus. Though this depiction only presents two coupling placements **503**, an additional coupling placement **503c** is present on target frame **501** in a position that is horizontally symmetrical to coupling placement **503a**. Some embodiments may comprise additional or different arrangements of coupling placements **503** without deviating from the teachings disclosed herein. Some embodiments may comprise fewer coupling placements **503** without deviating from the teachings disclosed herein. Some embodiments may comprise a guide arm **115** that may not be detached from target frame **501** without deviating from the teachings disclosed herein.

(19) FIG. 6 comprises a closeup illustration of the coupling components of the reference structure **105** (see FIG. 1; FIG. 2) and offset-target structure **113** (see FIG. 1; FIG. 2). Guide arm **115** comprises a sleeve **601** configured to receive a portion of horizontal member **107** moving in direction **609** during coupling. Sleeve **601** comprises a number of sleeve boundaries **603**, and a guide arm retainer **607**. Sleeve **601** is configured such horizontal member **107** fits snugly into sleeve **601**. With a snug fit, the motion of both horizontal member **107** and guide arm **115** with respect to one another is restricted in linear directions. Sleeve boundaries **603a** and **603b** restrict the vertical displacement of the coupled components. Sleeve boundary **603d** restricts the motion of the components with respect to direction **609**. Sleeve boundary **603b** and guide arm retainer **607** restrict the motion of the components with respect to a horizontal displacement in a direction perpendicular to direction **609**. The coupling is made detachable because the components can be readily moved in a direction opposite of direction **609** when coupled.

(20) Sleeve **601** is configured to restrict rotational movement of the coupled components, and in particular restriction rotational movement of offset-target structure **113**. Sleeve boundaries **603a**, **603b**, and **603c**, coordinating with guide arm retainer **607** restrict the pitch and roll of guide arm **115** when coupled with horizontal member **107**. Sleeve boundary **603b** and guide arm retainer **607** restrict the yaw displacement of the coupled components. The design of guide arm retainer **607** may be adjusted to restrict the yaw motion to a greater or lesser degree while still permitting a desired level of detachability between the components during coupling. In the depicted embodiment, guide arm retainer **607** retains horizontal member for approximately one-fourth the total length of sleeve **601**, but other embodiments may have different configurations providing different levels of desired yaw restriction without deviating from the teachings disclosed herein. By way of example and not limitation, guide arm retainer **607** may comprise the entire length of sleeve **601** to maximize the ability of sleeve **601** to secure the coupling between horizontal member **107** and guide arm **115**. Other ones of the number of sleeve boundaries **603** may comprise different configurations to optimize the desired security of sleeve **601**. In some embodiments, sleeve **601** may comprise a latch, pin-and-loop-, lock, or other security mechanism to enhance the security of the interface between horizontal member **107** and guide arm **115** during detachable coupling without deviating from the teachings disclosed herein.

(21) Because the position of offset-target structure **113** is determined based upon the coupling thereof to reference structure **105**, the position of offset-target structure **113** is effectively rendered automatic by the coupling. Because the interface between horizontal member **107** and guide arm

115 restricts the angular and rotational placement of offset-target structure **113**, the orientation of the offset-target structure **113** is also effectively rendered automatic by the coupling. In this regard, the calibration apparatus effectively renders a proper placement of the offset-target structure **113** entirely based upon the pre-existing work done to position reference structure **105** at its respective reference locus, thus saving additional measurements and placement determination for the additional structure. This setup saves time and efficiently permits a calibration for all sensors of a subject vehicle, including offset sensors.

(22) FIG. 7 is a flowchart depicting a method for setting up a calibration apparatus for calibrating an offset sensor of a subject vehicle. The method starts at step **700** by moving the subject vehicle into a space suitable for calibration that can accommodate the space required for placement of the calibration apparatus structures. In some embodiments, such as field implementations, this step may be omitted without deviating from the teachings disclosed herein.

(23) Once the vehicle is determined to be in a suitable location for setup and calibration, the method proceeds to step **702**, where a wheel-target structure is placed onto a wheel of the subject vehicle. In some embodiments using a calibration apparatus that does not rely upon a wheel-target structure, this step may be omitted without deviating from the teachings disclosed herein.

(24) Once the vehicle is fully-prepared, a camera associated with a reference structure is activated, and the camera is used to generate image data depicting the subject vehicle in step **704**. The image data is compared on a display to a reference template indicating an expected set of image data when the reference structure is properly positioned at a reference locus specified for the make model of the subject vehicle or the associated sensor intended for a calibration. The display is updated in real-time as the reference structure is moved around in step **706** to find the reference locus. At step **708**, a comparison is made between the active image data and the reference template to determine if the reference structure has been positioned at the reference locus. If not, the method continues in a loop back to step **706** as the reference structure continues being repositioned. Once the reference locus is successfully found, the method continues to step **710**, where the position of the reference structure is locked into place. In some embodiments wherein the calibration apparatus does not comprise a mechanism to lock the position of the reference structure, this step may be omitted without deviating from the teachings disclosed herein.

(25) After the reference structure is positioned appropriately, the method proceeds to step **712** where an offset-target structure is coupled to the reference structure. By way of example, and not limitation, the offset-target structure couples to the reference structure by utilizing a guide arm that is designed to automatically position the offset-target structure in a locus suitable for calibration of an associated offset sensor of the subject vehicle. In such an embodiment, the guide arm may restrict motion of the offset-target structure with respect to the reference structure (and by proxy, with respect to the reference locus) such that an appropriate placement and orientation of the offset-target structure is achieved by way of the coupling.

(26) Once coupled, the method may proceed to step **714**, where the coupling of the offset-target structure to the reference structure may be secured. In some embodiments, this step may be omitted without deviating from the teachings disclosed herein.

(27) Once the offset-target structure has been appropriately positioned by way of coupling, the calibration apparatus has been setup appropriately, and the method may continue to step **716** where the sensors are calibrated. In some embodiments, only the offset sensors may be calibrated at step **716**, and other sensors not affiliated with the offset-target structure may be calibrated at an earlier stage in the method after step **708** without deviating from the teachings disclosed herein.

(28) In some embodiments, the configuration of the calibration apparatus may necessitate additional intermediary steps not depicted in this embodiment without deviating from the teachings disclosed herein. Such intermediary steps may comprise attachment of a detachable target to the wheel-target structure, reference structure, or offset-target structure. Such intermediary steps may comprise placement of the guide arm to a particular coupling placement on the offset-target

structure. Such intermediary steps may comprise configuration or arrangement of the components of the wheel-target structure, reference structure, or offset-target structure. Such intermediary steps may comprise setup of the display to properly reflect a particular combination of vehicle model and sensor model(s) intended for calibration. Other steps of the method may be present without deviating from the teachings disclosed herein.

(29) While exemplary embodiments, are described above, it is not intended that these embodiments describe all possible forms of the disclosed apparatus and method. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the disclosure as claimed. The features of various implementing embodiments may be combined to form further embodiments of the disclosed concepts.

Claims

1. A calibration apparatus for an offset sensor of a subject vehicle, the apparatus comprising: a reference structure having a horizontal member, the horizontal member disposed in a direction parallel with a horizontal plane; a camera configured to be coupled to the reference structure and operable to generate image data depicting the subject vehicle; and an offset-target structure having a guide arm and configured to be detachably coupled to the horizontal member such that the guide arm interfaces with the horizontal member, wherein the calibration apparatus is configured for a calibration procedure of the offset sensor when the reference structure is positioned relative to the subject vehicle based upon image data generated by the camera and the offset-target structure is coupled to the horizontal member.
2. The calibration apparatus of claim 1, wherein the guide arm restricts the movement of the offset-target structure while the offset-target structure is coupled to the horizontal member.
3. The calibration apparatus of claim 1, wherein the guide arm restricts the pitch, yaw, or roll of the offset-target structure while the offset-target structure is coupled to the horizontal member.
4. The calibration apparatus of claim 1, wherein the guide arm restricts the pitch, yaw, and roll of the offset-target structure while the offset-target structure is coupled to the guide arm.
5. The calibration apparatus of claim 1, wherein the guide arm comprises a sleeve configured to receive a portion of the horizontal member when the guide arm interfaces with the horizontal member.
6. The calibration apparatus of claim 1, wherein the guide arm further comprises a retainer, the retainer configured to secure the guide arm and the horizontal member during coupling.
7. The calibration apparatus of claim 1, wherein the guide arm is configured to be detachably coupled to a number of coupling placements of the offset-target structure, the coupling placements comprising at least a first position suitable for coupling the offset-target structure to the horizontal member and a second position suitable for storing the guide arm.
8. The calibration apparatus of claim 1, further comprising a processor in data communication with the camera, the processor configured to compare a silhouette reference template in comparison with the image data and output guidance for placement of the reference structure with respect to the subject vehicle.
9. The calibration apparatus of claim 1, wherein the offset-target structure further comprises a detachable target.
10. The calibration apparatus of claim 9, wherein the detachable target comprises a vinyl mat target.
11. A method for placing an offset-target structure into a locus suitable for calibration of an offset sensor of a subject vehicle, the offset-target structure having a guide arm, the method comprising: positioning a reference structure into a reference locus, the reference structure having a horizontal member and a camera, the reference locus defined in relation to the vehicle and found using image

data generated by the camera depicting the subject vehicle; and coupling the offset-target structure to the reference structure such that the guide arm interfaces with the horizontal member, wherein the reference locus is found using image data generated by the camera and wherein the guide arm restricts the motion of the offset-target structure with respect to pitch, yaw, or roll when the offset-target structure is coupled to the reference structure.

12. The method of claim 11, wherein the guide arm comprises a sleeve, and the guide arm interfaces with the horizontal member by receiving a portion of the horizontal member within the sleeve.

13. The method of claim 12, wherein the guide arm further comprises a retainer, and wherein coupling the offset-target structure to the reference structure comprises the retainer securing the portion of the horizontal member received by the sleeve.

14. The method of claim 11, wherein the step of positioning the reference structure further comprises: generating the image data using the camera; outputting the image data to a display; and positioning the reference structure such that the image data on the display is aligned with a silhouette reference template superimposed on the display, the silhouette reference template illustrating expected image data generated when the reference structure is positioned at the reference locus.

15. The method of claim 11, further comprising coupling a detachable target to the offset-target structure, the detachable target having a design specified for calibration of the offset sensor.

16. The method of claim 11, wherein the coupling restricts the motion of the offset-target structure with respect to pitch, yaw, and roll.
